

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
3	(a)	(i)	Use of Newton's gravitational force equation (1) Answer = 1.266×10^{20} [N] or final substitution seen i.e. $\frac{6.67 \times 10^{-11} \times (6.2 \times 10^{25})^2}{(4.5 \times 10^{10})^2}$ (1)	1	1		2	2	
		(ii)	Each planet experiences two [equal and attractive] forces [from each of the other two] (1) N.B. same forces can be implied Resultant must be along centre lines (from symmetry) (1) $\times \cos 30^\circ$ seen (1) $2 \times 1.3 \times 10^{20} \times \cos 30^\circ$ (seen) OR 2.25 OR 2.19 (seen) (1)		4		4	2	
	(b)	(i)	30° angle seen or explained (1) $\cos 30^\circ = \frac{2.25 \times 10^{10}}{d}$ or equivalent seen (1)		2		2	2	
		(ii)	Mass is [far] greater (1) can be implied if substituted Factor of 10^5 or calculation 2.5×10^{25} N so Rhodri is correct (1) No penalty for incorrect distance as long as force is much bigger			2	2	1	
	(c)	(i)	Potential equation used i.e. $[-] \frac{GM}{r}$ (1) Potential due to star = $[-] 1.05 \times 10^{10}$ (1) Other potentials negligible (either stated or full calculation) (1)	1	1 1		3	2	

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		(ii)	PE of system / planet = $[-]1 \times 10^{10} \times 6.2 \times 10^{25} = (6.2 \times 10^{35} \text{ J})$ (1) Alternative for the 1st mark: Obtaining escape velocity: $140\,000 \text{ [m s}^{-1}\text{]}$ and velocity of $7 \times 10^{35} \text{ [J]}$ i.e. $150\,000 \text{ [m s}^{-1}\text{]}$ Another alternative for the 1st mark: $\frac{7 \times 10^{35}}{6.2 \times 10^{25}} = 1.13 \times 10^{10} \text{ [J kg}^{-1}\text{]}$ However, conservation of momentum / other energy transfers must be considered (1) minimum required is {elastic / inelastic} collision Conclusion consistent with their argument but must be based on a value of PE or potential or escape velocity (1)			3	3	1	
	(d)		Star is at centre of mass or resultant force will be zero (1) Hence, no wobble / orbit etc. or stationary (1) No Doppler shift and not detectable so Delyth is correct (1)			3	3		
			Question 3 total	2	9	8	19	10	0